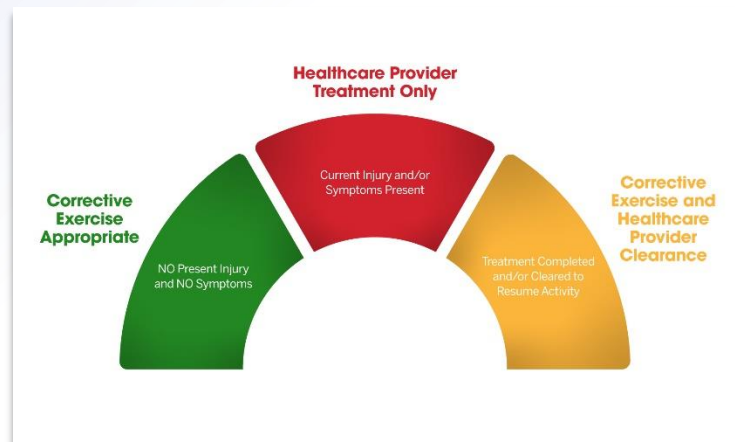


Use this study guide with the online course and textbook to review essential knowledge topics and prepare for your exam.

Scope of Practice

Scope of practice – Procedures and actions professionals are permitted to perform within the boundaries of their profession

Given the scope of fitness professionals, it is important to recognize where a client is in their recovery from injury, referring to another professional, as necessary.



Biomechanical and Anatomy Key Terms

Agonist – The prime mover muscle for a given movement pattern or joint action

Antagonist – A muscle that acts in direct opposition to the prime mover

Concentric muscle action – Occurs when a muscle generates force while shortening to accelerate an external load

Eccentric muscle action – Occurs when a muscle generates force while lengthening to decelerate an external load

Isometric muscle action – Occurs when a muscle generates force equal to an external load to hold it in place

Kyphosis – Natural curvature of the thoracic spine toward the back of the body

Length-tension relationship – The resting length of a muscle and the tension the muscle can produce at this resting length

Lordosis – Natural curvature of the lumbar or cervical spine toward the front of the body

Motor behavior – The human movement system's response to internal and external environmental stimuli

Motor control – The study of posture and movements with the involved structures and mechanisms used by the central nervous system to assimilate and integrate sensory information with previous experiences

Motor development – The change in motor behavior over time throughout a person's life span

Motor learning – The utilization of these processes through practice and experience leading to a relatively permanent change in a person's capacity to produce skilled movements

Movement compensation – When the body moves in a suboptimal way in response to kinetic chain dysfunction

Neural drive – The rate and volume of activation signals a muscle receives from the central nervous system.

Neuromuscular efficiency – The ability of the neuromuscular system to allow agonist, antagonists, synergists, and stabilizers to work synergistically to produce, reduce, and dynamically stabilize the Human Movement System in all three planes of motion

Overactive/shortened – Occurs when elevated neural drive causes a muscle to be held in a chronic state of contraction

Proprioception – The cumulative neural input from sensory afferents to the central nervous system

Reciprocal inhibition – When an agonist contracts, its functional antagonist relaxes to allow movement to occur at a joint.

Synergist – Muscles that assist prime movers during functional movement patterns

Stabilizer – Muscles that support or stabilize the body while the prime movers and the synergists perform the movement patterns

Underactive/lengthened – Occurs when inhibited neural drive allows a muscle's functional antagonist to pull it into a chronically elongated state

The Regional Interdependence Model (RI)

Regional interdependence (RI) model – The concept that impairments in one musculoskeletal region will influence the movement quality and functional capacity of others

Local Muscular System

System	Function	Muscles
Local Muscular System	Responsible for stabilization and used for endurance, balance, and slow movement training	▪ Transversus abdominis ▪ Multifidus ▪ Internal oblique ▪ Psoas ▪ Diaphragm ▪ Muscles of the pelvic floor



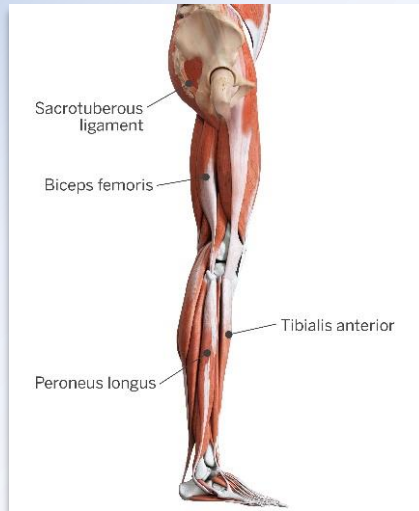
Global Muscular System

System	Function	Muscles
Global Muscular System	Responsible for movement and used for strength, coordination, agility, and fast velocity training	▪ Rectus abdominus ▪ External obliques ▪ Erector spinae ▪ Hamstring complex ▪ Gluteus maximus ▪ Latissimus dorsi ▪ Adductors ▪ Quadriceps ▪ Gastrocnemius

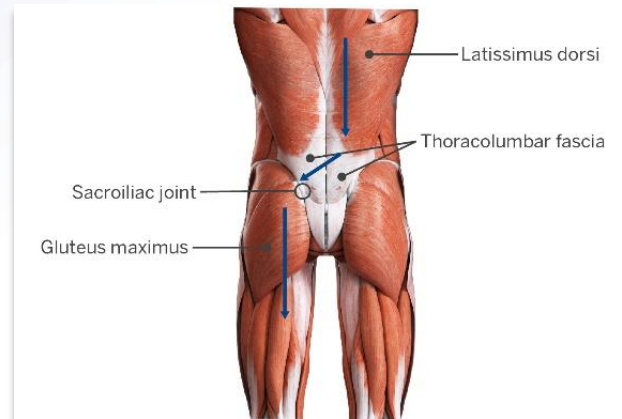


Subsystems of the Global Muscular System

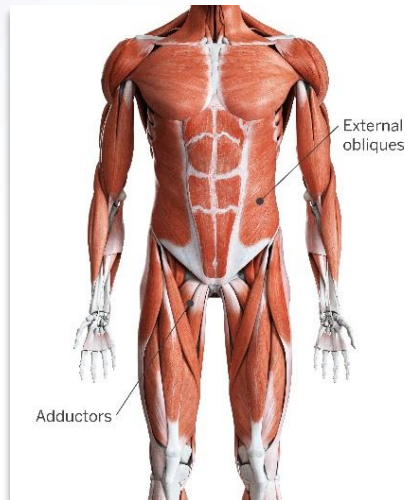
Deep Longitudinal Subsystem (DLS)



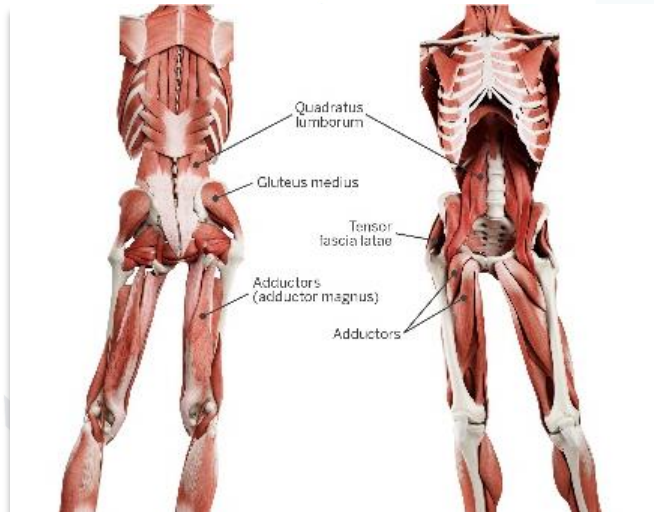
Posterior Oblique Subsystem (POS)



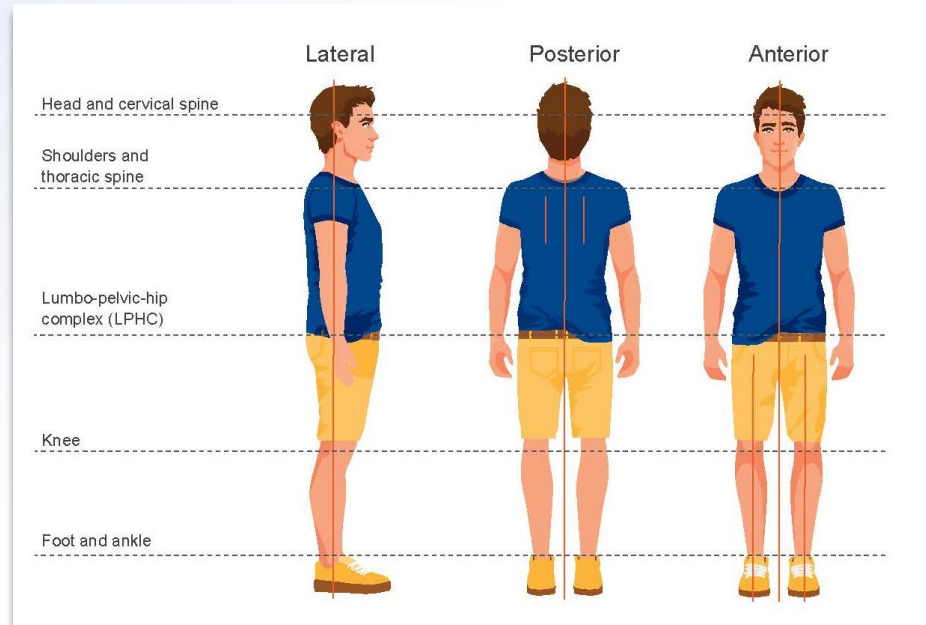
Anterior Oblique Subsystem (AOS)



Lateral Subsystem (LS)



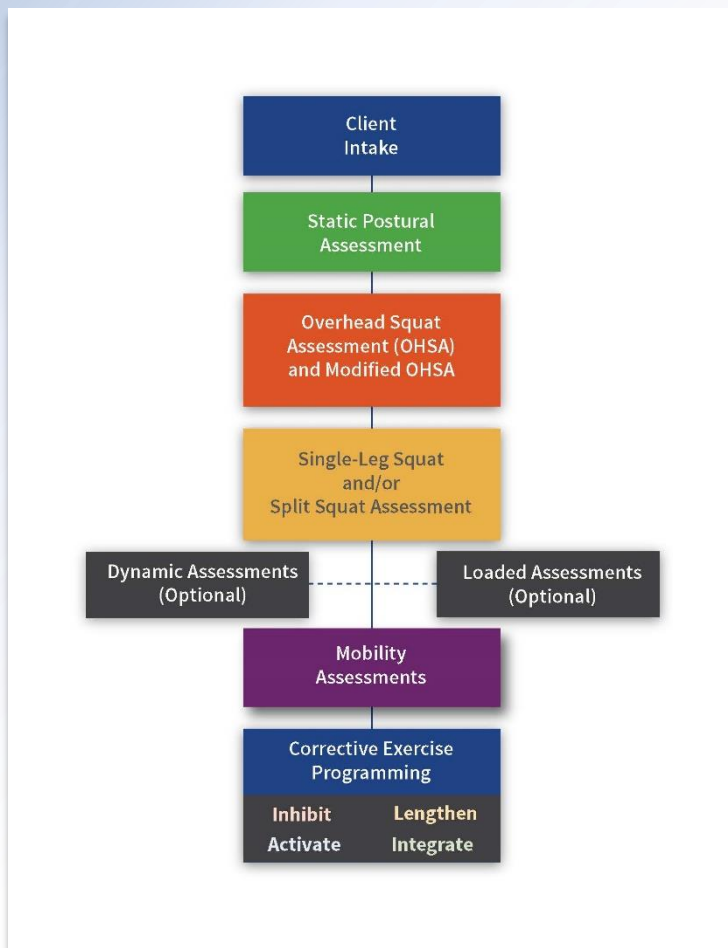
Five Kinetic Chain Checkpoints



Corrective Exercise Continuum



CES Assessment Flow



Precautions and Contraindications

Self-Myofascial Rolling

Precautions		
- Hypertension (controlled) - Diabetes - Abnormal sensations (e.g., numbness) - Inability to position the body or perform myofascial rolling correctly	- Osteopenia - Varicose veins - Sensitivity to pressure - Young children - Pregnancy	- Bony prominences or regions - Recent injury or surgery - Elderly

Contraindications (Self-Myofascial Rolling)		
▪ Skin rash, open wounds, blisters, local tissue inflammation, bruises, or tumors ▪ Bone fracture of myositis ossificans ▪ Acute or severe cardiac, liver, or kidney disease ▪ Bleeding disorders ▪ Connective tissue disorders ▪ Direct pressure over surgical site or hardware ▪ Pregnancy (consult MD) ▪ Severe scoliosis or spinal deformity	▪ Deep vein thrombosis ▪ Cancer or malignancy ▪ Acute infection (viral or bacterial), fever, or contagious condition ▪ Systemic conditions (e.g., diabetes) ▪ Peripheral vascular insufficiency or disease ▪ Chronic pain conditions (e.g., rheumatoid arthritis) ▪ Direct pressure over bony prominences or regions (e.g., lumbar vertebrae) ▪ Osteomyelitis	▪ Osteoporosis ▪ Hypertension (uncontrolled) ▪ Neurologic conditions resulting in loss or altered sensation ▪ Recent surgery or injury ▪ Medications that thin the blood or alter sensations ▪ Direct pressure over face, eyes, arteries, veins (e.g., varicose veins), or nerves ▪ Extreme discomfort or pain felt by client

Stretching, Isolated Strengthening, and Integrated Dynamic Movement

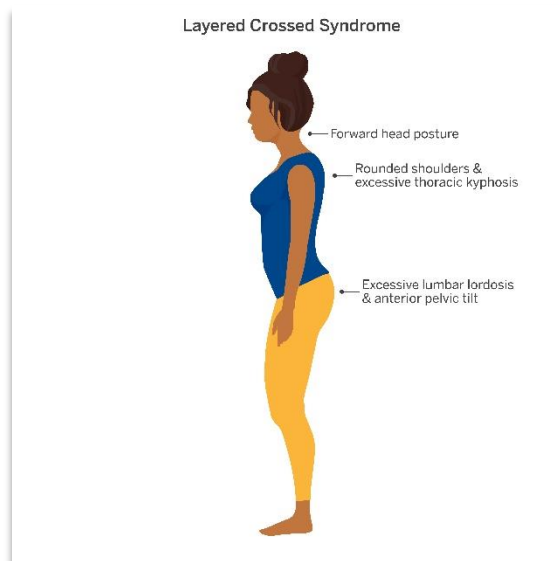
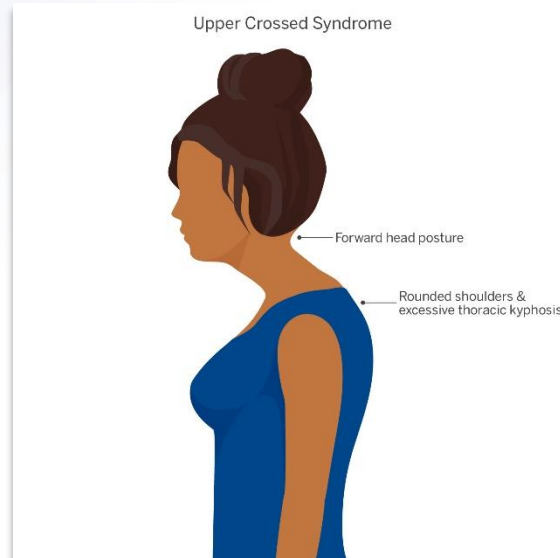
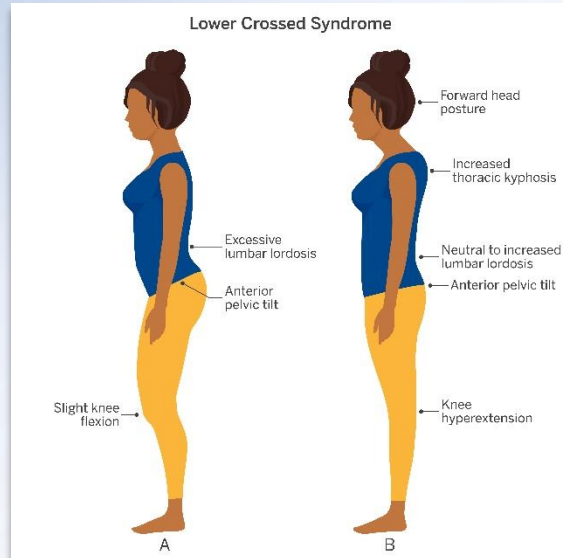
Stretching	
Precautions	Contraindications
▪ Special populations (e.g., pregnant women, osteoarthritis, and rheumatoid arthritis) ▪ Seniors ▪ Hypertensive patients ▪ Neuromuscular disorders ▪ Joint replacements ▪ Fibromyalgia ▪ Marfan syndrome	▪ Acute injury or muscle strain or tear of the muscle being stretched ▪ Recent musculoskeletal surgery or treatment (i.e., shoulder dislocations, ligament repairs, or fractures) ▪ Acute rheumatoid arthritis of the affected joint ▪ Osteoporosis (NMS)
Isolated Strengthening	
Precautions	Contraindications
▪ Special populations ▪ Neuromuscular disorders ▪ Clients with poor core stabilization strength	▪ Acute injury or muscle strain or tear of the muscle being strengthened ▪ Acute rheumatoid arthritis of the affected joint ▪ Impaired joint motion ▪ Pain produced during the movement
Integrated Dynamic Movement	
Precautions	Contraindications
▪ Special populations ▪ Neuromuscular disorders	▪ Acute injury or muscle strain or tear of the muscle being worked ▪ Acute rheumatoid arthritis of the affected joint ▪ Position of exercise (prone, supine, or decline position) relative to the client's condition (pregnancy, coronary heart disease, etc.) ▪ Acute injury to joint involved during movement ▪ Pain

Acute Training Variables

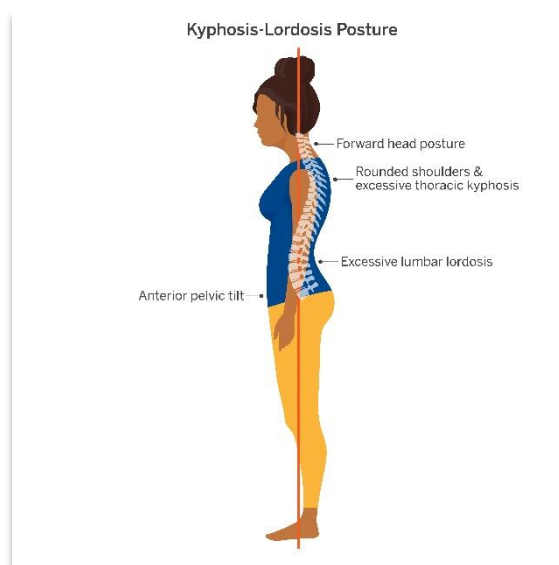
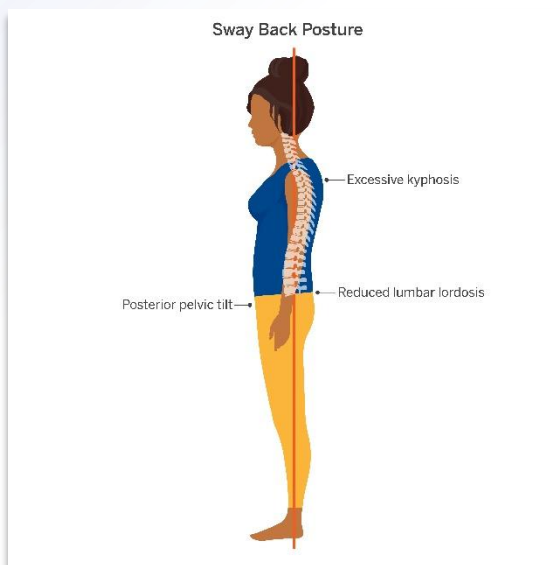
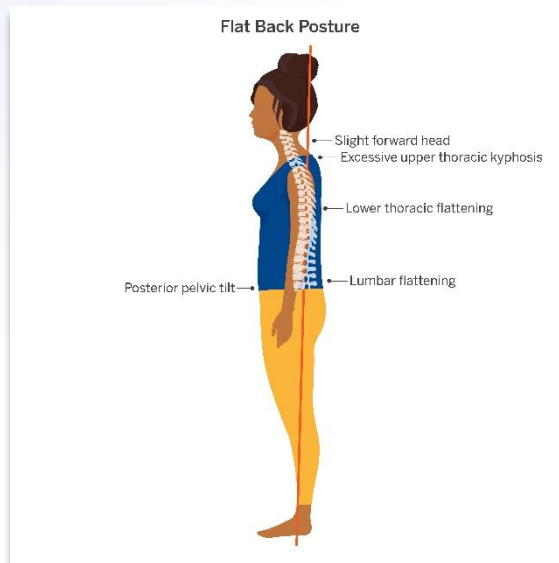
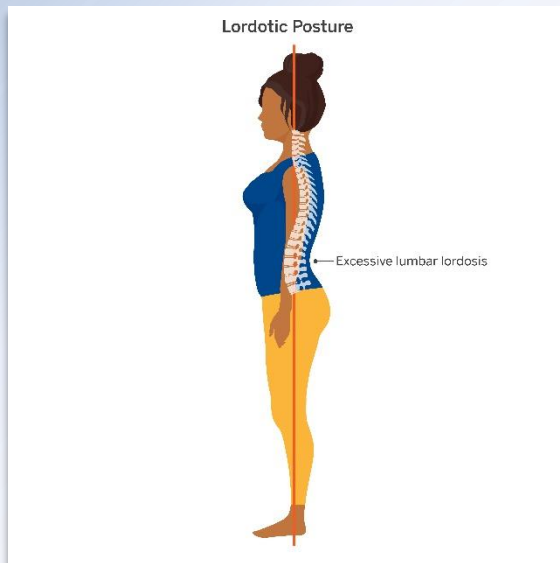
Acute Training Variables for Self-Myofascial Rolling – Inhibit				
Frequency	Sets	Repetitions	Intensity	Duration
Most days of the week (unless otherwise specified)	1	▪ Hold areas of discomfort for 30 to 60 seconds ▪ Perform four to six repetitions of active movement	Should be some discomfort, but able to relax and breathe	5 to 10 minutes total time; 90 to 120 seconds per muscle group
Acute Training Variables for Static Stretching – Lengthen				
Frequency	Sets	Repetitions	Duration	
Daily (unless specified otherwise)	n/a	1 to 4*	▪ 20- to 30-second hold ▪ 60-second hold for older clients (≥ 65 years)	
*Perform no more than 60 seconds of static stretching per muscle group if completed before an athletic competition or high-intensity activity.				
Acute Training Variables for Isolated Strengthening – Activate				
Frequency	Sets	Repetitions	Duration of Repetition	
3 to 5 days per week	1 to 2	10 to 15	▪ 4/2/1 ▪ 4 seconds eccentric ▪ 2 seconds isometric hold at end-range ▪ 1 second concentric	
Acute Training Variables for Integrated Dynamic Movement – Integrate				
Frequency	Sets	Repetitions	Duration of Repetition	
3 to 5 days per week	1 to 3	10 to 15	Controlled	

Common Patterns of Postural Distortion

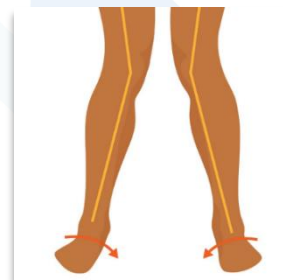
Janda's Postural Distortion Syndromes



Kendall's Posture Types



Pes planus distortion syndrome – A combination of excessive pes planus (flat feet), knee flexion (reduced knee extension ROM), hip and knee internal rotation, knee valgus (knock-kneed), and a pelvic anterior tilt.

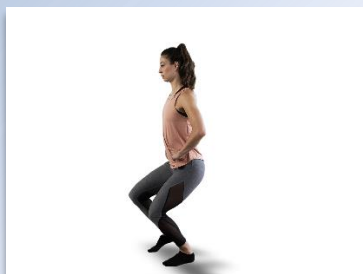


Common Movement Impairments

Note that each of the following movement impairments may be caused by, or associated with, other movement impairments due to the concepts of regional interdependence.

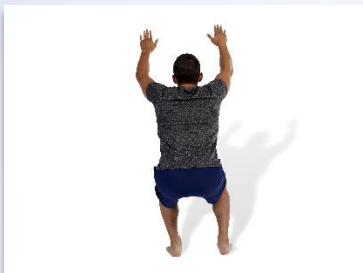
Common Movement Impairments		
	Excessive pronation	Look for the arch of the foot to collapse and flatten, eversion of the heel, or malalignment of the Achilles tendon.
	Feet turn out	Look for the toes to rotate laterally during the movement (also known as foot abduction).
	Heel rise	Look for the heel to come off of the ground during the movement.
	Knee valgus	Look for the knees to collapse inward.
	Knee varus	Look for the knees to bow outward.

Common Movement Impairments (Continued)



Knee dominance

Look for an upright trunk, the knees to move in front of the toes, and/or for more knee anterior displacement compared to hip posterior displacement; that is, the knees move forward more than the hips move back. May be seen with heel rise.



Asymmetric weight shift

Look for the hip to shift toward one side or the other. The side of the body opposite of the shift may also exhibit the hip dropping in the frontal plane.



Excessive trunk movement

Look for instability of the trunk when in a push-up position (*specifically during the dynamic Davies test*).



Excessive anterior pelvic tilt

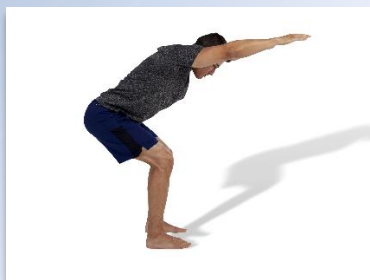
Look for the pelvis to roll forward and for the lumbar spine to extend beyond normal curvature, creating a prominent low-back arch.



Excessive posterior pelvic tilt

Look for the pelvis to roll backward and for the lumbar spine to flex, creating a flattening of the lower back.

Common Movement Impairments (Continued)



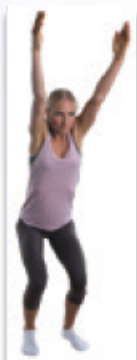
Excessive forward trunk lean

Look for the trunk to lean forward and beyond ideal parallel alignment with the shins.



Trunk rotation

Look for the trunk of the body to rotate internally or externally during single-leg movements.



Scapular elevation

Look for the shoulders to move up toward the ears.



Scapular winging

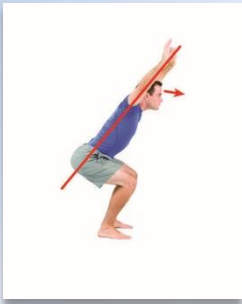
Look for the scapulae to protrude excessively from the back, seen most prominently in a push-up position (*specifically during the dynamic Davies test or when pushing or pulling*).



Arms fall forward

Look for the arms to fall forward to no longer be aligned with the torso and ears.

Common Movement Impairments (Continued)



Excessive cervical extension (forward head)

Look for the head to migrate forward, moving the ears out of alignment with the shoulders.

Movement Assessment Solutions

Checkpoint	View	Movement Impairment	Potential Contributors	Suggested Mobility Assessments*
Foot and Ankle	Anterior	Feet turn out	Overactive/shortened - Biceps femoris (short head) - Gastrocnemius (lateral) - Soleus Underactive/lengthened - Anterior tibialis - Gastrocnemius (medial) - Gluteus maximus - Gluteus medius - Hamstrings complex (medial) - Posterior tibialis	- Active knee extension - Ankle dorsiflexion - Hip abduction and external rotation - Modified Thomas test - Seated hip internal and external rotation
	Lateral	Heel rise	Overactive/shortened - Quadriceps complex - Soleus Underactive/lengthened - Anterior tibialis - Gluteus maximus	- Active knee flexion - Ankle dorsiflexion
	Posterior	Excessive pronation	Overactive/shortened - Fibularis (peroneal) complex - Gastrocnemius (lateral) - TFL Underactive/lengthened - Anterior tibialis - Gastrocnemius (medial) - Gluteus maximus - Gluteus medius - Intrinsic foot muscles - Posterior tibialis	- Ankle dorsiflexion - Modified Thomas test - Seated hip internal and external rotation

Checkpoint	View	Movement Impairment	Potential Contributors	Suggested Mobility Assessments*
Knee	Anterior	Valgus (inward)	Overactive/shortened - Adductor complex - Biceps femoris (short head) - Gastrocnemius - Soleus - TFL - Vastus lateralis Underactive/lengthened - Anterior tibialis - Gluteus maximus - Gluteus medius - Hamstrings complex (medial) - Posterior tibialis - Vastus medialis oblique (VMO)	- Active knee extension - Ankle dorsiflexion - Hip abduction and external rotation - Modified Thomas test - Seated hip internal and external rotation
Knee (continued)		Varus (outward)	Overactive/shortened - Adductor magnus (posterior fibers) - Anterior tibialis - Biceps femoris (long head) - Piriformis - Posterior tibialis - TFL Underactive/lengthened - Adductor complex - Gluteus maximus - Hamstrings complex (medial)	- Active knee extension - Lumbar flexion - Modified Thomas test - Passive hip internal rotation - Seated hip internal and external rotation
	Lateral	Knee dominance	Overactive/shortened [^] - Adductor magnus - Piriformis - Quadriceps complex - Soleus Underactive/lengthened [^] - Core stabilizers - Gluteus maximus	- Active knee flexion - Ankle dorsiflexion - Hip abduction and external rotation - Modified Thomas test - Passive hip internal rotation

Checkpoint	View	Movement Impairment	Potential Contributors	Suggested Mobility Assessments*
LPHC	Anterior or Posterior	Asymmetric weight shift	Overactive/shortened - Same side as shift - Adductor complex - TFL - Opposite side of shift - Biceps femoris - Gastrocnemius/soleus - Piriformis Underactive/lengthened - Core stabilizers - Same side as shift - Gluteus medius - Opposite side of shift - Adductor complex	- Active knee extension - Ankle dorsiflexion - Hip abduction and external rotation - Modified Thomas test - Seated hip internal and external rotation
		Excessive trunk movement during testing (Davies test)	Overactive/shortened - N/A Underactive/lengthened - Local core stabilizers	N/A
	Lateral	Excessive anterior pelvic tilt (increased lumbar extension)	Overactive/shortened - Adductor complex (anterior fibers) - Latissimus dorsi - Psoas - Rectus femoris - Spinal extensor complex (erector spinae and quadratus lumborum) - TFL Underactive/lengthened - External obliques - Gluteus maximus - Hamstrings complex - Local core stabilizers - Rectus abdominis	- Active knee flexion - Hip abduction and external rotation - Lumbar flexion and extension - Modified Thomas test - Shoulder flexion

Checkpoint	View	Movement Impairment	Potential Contributors	Suggested Mobility Assessments*
LPHC (continued)	Lateral	Excessive posterior pelvic tilt (increased lumbar flexion)	Overactive/shortened - Adductor magnus - External obliques - Hamstrings complex - Piriformis - Rectus abdominis Underactive/lengthened - Gluteus maximus - Latissimus dorsi - Local core stabilizers - Psoas - Rectus femoris - Spinal extensor complex (erector spinae and quadratus lumborum) - TFL	- Active knee extension - Hip abduction and external rotation - Lumbar flexion and extension - Seated hip internal and external rotation
		Excessive forward trunk lean	Overactive/shortened - Adductor complex (anterior fibers) - External obliques (if observed with lumbar flexion) - Gastrocnemius - Psoas - Rectus abdominis (if observed with lumbar flexion) - Rectus femoris - Soleus - TFL Underactive/lengthened - Anterior tibialis - Gluteus maximus - Hamstrings complex - Local core stabilizers - Spinal extensor complex (erector spinae and quadratus lumborum)	- Active knee flexion - Ankle dorsiflexion - Modified Thomas test
	Anterior	Inward trunk rotation (single-leg and split squat)	Overactive/shortened - Adductor complex - TFL Underactive/lengthened - Gluteus maximus - Gluteus medius - Local core stabilizers	- Hip abduction and external rotation - Modified Thomas test - Seated hip internal and external rotation

Checkpoint	View	Movement Impairment	Potential Contributors	Suggested Mobility Assessments*
LPHC (continued)	Anterior	Outward trunk rotation (single-leg and split squat)	Overactive/shortened - Adductor magnus (posterior fibers) - Hamstrings complex (lateral) - Piriformis Underactive/lengthened - Adductor complex (anterior fibers) - Gluteus maximus - Gluteus medius - Local core stabilizers	- Hip abduction and external rotation - Modified Thomas test - Seated hip internal and external rotation
Shoulders and Thoracic Spine	Anterior or Posterior	Scapular elevation	Overactive/shortened - Levator scapulae - Pectoralis minor - Upper trapezius Underactive/lengthened - Lower trapezius - Serratus anterior	- Cervical flexion and extension - Cervical lateral flexion - Cervical rotation - Seated thoracic rotation - Thoracic extension
	Lateral	Scapular winging (Davies test and push assessment)	Overactive/shortened - Latissimus dorsi - Pectoralis minor - Upper trapezius Underactive/lengthened - Lower trapezius - Middle trapezius - Serratus anterior	- Seated thoracic rotation - Shoulder flexion - Shoulder retraction - Thoracic extension
		Arms fall forward	Overactive/shortened - Latissimus dorsi - Pectoralis major - Pectoralis minor - Teres major Underactive/lengthened - Infraspinatus - Lower trapezius - Middle trapezius - Posterior deltoids - Rhomboids - Teres minor	- Cervical flexion and extension - Cervical rotation - Cervical lateral flexion - Shoulder extension - Shoulder flexion - Shoulder internal and external rotation - Shoulder retraction - Seated thoracic rotation - Thoracic extension

Checkpoint	View	Movement Impairment	Potential Contributors	Suggested Mobility Assessments*
Head and Cervical Spine	Lateral	Excessive cervical extension (forward head)	Overactive/shortened ▪ Cervical extensors (suboccipital) ▪ Levator scapulae ▪ Sternocleidomastoid ▪ Upper trapezius Underactive/lengthened ▪ Deep cervical flexors ▪ Lower trapezius ▪ Middle trapezius ▪ Rhomboids	Cervical flexion and extension Cervical lateral flexion Cervical rotation
*It is not necessary to perform all of the listed mobility assessments associated with each movement impairment. The mobility assessments provided are a starting point that is narrowed down based on the results of the OHSA, Modified OHSA, and other movement assessments. It is likely that only a few mobility assessments will be needed.				
^Movement competency, pain avoidance, or balance strategies should be ruled out prior to assuming over- and underactive muscles as contributing factors to knee dominance.				

Mobility Assessments

How Results Influence Programming

Mobility assessments help direct exercise programming toward flexibility or strengthening strategies for that impairment.

Training Tip

If the client demonstrates a specific, noticeable restriction of mobility, then it is recommended to inhibit and lengthen (Phases 1 and 2 of the Corrective Exercise Continuum) muscles identified as potentially overactive/shortened. However, if a client demonstrates optimal mobility in these tests, then emphasis should be placed on activating muscles that were identified as potentially underactive/lengthened during the previous static and movement assessments.

Helpful Hint

The following is a helpful example that illustrates how mobility assessments refine programming.

Knee valgus observed → hip adductors potentially overactive/shortened → assess adductor mobility by observing abduction (the opposite motion)

→ if restricted → overactive/shortened hip adductors contribute to knee valgus → program inhibition and lengthening of the hip adductors

→ if mobility is normal → underactive/lengthened hip abductors contribute to knee valgus → program isolated strengthening of the hip abductors

Overactive Muscles Associated With Restriction in Each Assessment

Mobility Assessment	Overactive/Shortened Muscle(s)
Ankle Dorsiflexion (Weight-Bearing Lunge Test)	Gastrocnemius and soleus
First MTP (Great Toe) Extension	Flexor hallucis longus
Knee Flexion Test (Duncan-Ely Test)	Quadriceps complex
Active Knee Extension Test	Hamstrings complex
Lumbar Flexion	Erector spinae
Lumbar Extension	Rectus abdominus, internal obliques, external obliques
Hip Extension, Hip Adduction, and Knee Flexion (Modified Thomas Test)	▪ Hip extension: Psoas and rectus femoris ▪ Hip adduction: Tensor fasciae latae ▪ Knee flexion: Rectus femoris
Hip Abduction and External Rotation (Adductor Test)	Hip adductor complex
Passive Hip Internal Rotation	Piriformis, quadratus femoris, and gluteus maximus
Seated Hip Internal Rotation	Piriformis, gemellus superior, gemellus inferior, obturator internus, obturator externus, quadratus femoris, and gluteus maximus
Seated Hip External Rotation	Tensor fasciae latae, gluteus minimus and medius (anterior fibers), and hip adductors
Shoulder Flexion (Lat Length Test)	Latissimus dorsi, teres major, and pectoralis major (lower fibers)
Shoulder Retraction (Pectoralis Minor Test)	Pectoralis minor on the same side as the elevated shoulder or compensation
Shoulder Extension	Anterior deltoid, pectoralis major (upper fibers), coracobrachialis, and biceps brachii
Shoulder Internal Rotation	Teres minor and infraspinatus
Shoulder External Rotation	Subscapularis, teres major, latissimus dorsi, and pectoralis major
Elbow Flexion	Triceps group
Elbow Extension	Biceps brachii, brachialis, brachioradialis, and pronator teres
Wrist Flexion	Wrist extensors (extensor carpi radialis longus, extensor carpi radialis brevis, and extensor carpi ulnaris)
Wrist Extension	Wrist flexors (flexor carpi radialis, flexor carpi ulnaris, and palmaris longus)
Cervical Flexion	Erector spinae, deep cervical extensors, and upper trapezius
Cervical Extension	Sternocleidomastoid and deep cervical flexors
Cervical Rotation	Sternocleidomastoid and scalenes on the side opposite of the observed restriction
Cervical Side Bending (Lateral Flexion)	Sternocleidomastoid, scalenes, and erector spinae on the side opposite of the observed restriction
Thoracic Extension	Rectus abdominis, internal oblique, and external oblique
Seated Thoracic Rotation	Rectus abdominis, internal oblique, external oblique, and erector spinae on the side opposite of the restriction

Common Corrective Exercise Programming Selections

Common Corrective Exercise Programming Selections for the Foot and Ankle

Phase	Modality	Muscle(s)/Exercise	Acute Training Variables
Inhibit	Self-myofascial rolling	- Biceps femoris (short head) - Fibularis complex (peroneals) - Gastrocnemius - Quadriceps - Soleus - TFL	- Hold areas of discomfort for 30 to 60 seconds. - Perform four to six repetitions of active joint movement.
Lengthen	Static or neuromuscular stretching (NMS)	- Biceps femoris (short head) - Gastrocnemius - Quadriceps - Soleus - TFL	- Static: 30-second hold - NMS: 7- to 10-second isometric contraction, 30-second static hold
Activate	Isolated strengthening	- Anterior tibialis - Gluteus medius - Medial hamstrings - Posterior tibialis - Short foot (intrinsic muscles)	10 to 15 reps with 4-second eccentric contraction, 2-second isometric contraction at end-range, and 1-second concentric contraction
Integrate*	Integrated dynamic movement	- Single-leg balance reach - Step-up to balance - Lunge to balance progressions - Single-leg squat	10 to 15 reps under control

*NOTE: Progress and regress as needed to match client ability, work capacity, and needs.

Common Corrective Exercise Programming Selections for the Knee

Phase	Modality	Muscle(s)/Exercise	Acute Training Variables
Inhibit	Self-myofascial rolling	- Adductor complex - Biceps femoris - Fibularis complex (peroneals) - Gastrocnemius - Piriformis - Quadriceps - Soleus - TFL	- Hold areas of discomfort for 30 to 60 seconds - Perform four to six repetitions of active joint movement.

Common Corrective Exercise Programming Selections for the Knee (Continued)			
Phase	Modality	Muscle(s)/Exercise	Acute Training Variables
Lengthen	Static or neuromuscular stretching (NMS)	- Adductor complex (for valgus) - Biceps femoris - Gastrocnemius - Hip flexor complex - Piriformis - Quadriceps - Soleus - TFL	- Static: 30-second hold - NMS: 7- to 10-second isometric contraction, 30-second static hold
Activate	Isolated strengthening	- Adductor complex (for varus) - Anterior tibialis - Core stabilizers - Gluteus maximus - Gluteus medius - Medial hamstrings - Posterior tibialis	10 to 15 reps with 4-second eccentric contraction, 2-second isometric contraction at end-range, and 1-second concentric contraction
Integrate*	Integrated dynamic movement	- Lateral tube walking - Lunge to balance progressions - Single-leg squat - Squat with medicine ball between knees (for varus) - Squat with mini-band around knees (for valgus) - Step-up to balance - Wall jump	10 to 15 reps under control
*NOTE: Progress and regress as needed to match client ability, work capacity, and needs.			

Common Corrective Exercise Programming Selections for the LPHC

Phase	Modality	Muscle(s)/Exercise	Acute Training Variables
Inhibit	Self-myofascial rolling	- Adductor complex - Adductor magnus - Biceps femoris - Gastrocnemius/soleus - Hamstrings complex - Hip flexor complex - Latissimus dorsi - Piriformis - Rectus femoris - TFL	- Hold areas of discomfort for 30 to 60 seconds. - Perform between four to six repetitions of active joint movement.

Common Corrective Exercise Programming Selections for the LPHC (Continued)			
Phase	Modality	Muscle(s)/Exercise	Acute Training Variables
Lengthen	Static or neuromuscular stretching (NMS)	▪ Abdominal complex ▪ Adductor complex ▪ Adductor magnus ▪ Biceps femoris ▪ Gastrocnemius/soleus ▪ Hamstrings complex ▪ Hip flexor complex ▪ Piriformis ▪ Spinal extensor complex ▪ TFL	▪ Static: 30-second hold ▪ NMS: 7- to 10-second isometric contraction, 30-second static hold
Activate	Isolated strengthening	▪ Adductor complex ▪ Anterior tibialis ▪ Core stabilizers ▪ Gluteus maximus ▪ Gluteus medius ▪ Hamstrings complex ▪ Hip flexor complex ▪ Latissimus dorsi ▪ Rectus abdominis ▪ Spinal extensor complex	10 to 15 reps with 4-second eccentric contraction, 2-second isometric contraction at end-range, and 1-second concentric contraction
Integrate*	Integrated dynamic movement	▪ Ball wall squat with overhead press ▪ Cable squat to row ▪ Lateral tube walking ▪ Lunge to overhead press ▪ Step up to overhead cable press	10 to 15 reps under control
*NOTE: Progress and regress as needed to match client ability, work capacity, and needs.			

Common Corrective Exercise Programming Selections for the Thoracic Spine and Shoulder

Phase	Modality	Muscle(s)/Exercise	Acute Training Variables
Inhibit	Self-myofascial rolling	▪ Biceps brachii ▪ Latissimus dorsi ▪ Levator scapulae ▪ Pectoralis major ▪ Pectoralis minor ▪ Thoracic spine ▪ Upper trapezius	▪ Hold areas of discomfort for 30 to 60 seconds. ▪ Perform between four to six repetitions of active joint movement.

Lengthen	Static or neuromuscular stretching (NMS)	▪ Biceps brachii ▪ Latissimus dorsi ▪ Levator scapulae ▪ Pectoralis major ▪ Pectoralis minor ▪ Posterior capsule/deltoid ▪ Upper trapezius	▪ Static: 30-second hold ▪ NMS: 7- to 10-second isometric contraction; 30-second hold
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Common Corrective Exercise Programming Selections for the Thoracic Spine and Shoulder (Continued)			
Phase	Modality	Muscle(s)/Exercise	Acute Training Variables
Activate	Isolated strengthening	▪ Ball combo 1 ▪ Ball combo 2 ▪ Cobra ▪ Push-up plus ▪ Rotator cuff (resisted internal and external rotation) ▪ Scaption	10 to 15 reps with 4-second eccentric contraction, 2-second isometric contraction at end-range, and 1-second concentric contraction
Integrate*	Integrated dynamic movement	▪ Pulling progressions ▪ Pushing progressions ▪ Single-leg RDL to PNF pattern ▪ Squat to row	10 to 15 reps under control
*NOTE: Progress and regress as needed to match client ability, work capacity, and needs.			

Common Corrective Programming Selections for the Elbow and Wrist

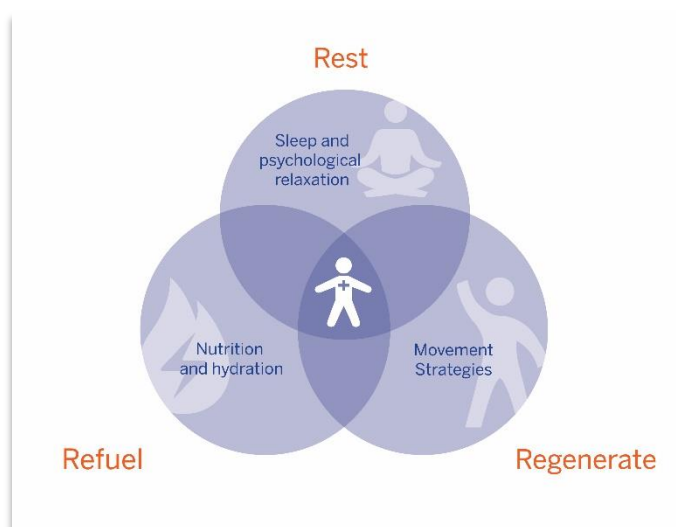
Phase	Modality	Muscle(s)/Exercise(s)	Acute Training Variables
Inhibit	Self-myofascial rolling (SMR) (using fingertips or massage ball)	▪ Biceps brachii ▪ Brachialis ▪ Wrist extensors ▪ Wrist flexors	▪ Hold areas of discomfort for 30 to 60 seconds. ▪ Perform between four to six repetitions of active joint movement.
Lengthen	Static stretching	▪ Biceps brachii ▪ Wrist extensors ▪ Wrist flexors	30-second hold
Activate	Isolated strengthening	▪ Elbow extension ▪ Wrist flexion or extension	10 to 15 reps with 4-second eccentric contraction, 2-second isometric contraction at end-range, and 1-second concentric contraction
Integrate*	Integrated dynamic movement	▪ Inverted row ▪ Prone ball triceps extension with cobra ▪ Standing cable press ▪ Triceps extension progressions	10 to 15 reps under control
*NOTE: Progress and regress as needed to match client ability, work capacity, and needs.			

Common Corrective Programming Selections for the Cervical Spine

Phase	Modality	*Muscle(s)/Exercise	Acute Training Variables
Inhibit	Self-myofascial rolling	▪ Cervical extensors (suboccipitals) ▪ Levator scapulae ▪ Sternocleidomastoid ▪ Thoracic spine ▪ Upper trapezius	▪ Hold areas of discomfort for 30 to 60 seconds. ▪ Perform between four to six repetitions of active joint movement.
Lengthen	Static stretching	▪ Levator scapulae ▪ Scalenes (included when stretching the upper trapezius and sternocleidomastoid) ▪ Sternocleidomastoid ▪ Upper trapezius	30-second hold
Activate	Isolated strengthening	▪ Cobra progressions ▪ Deep cervical flexors (chin-tuck progressions) ▪ Scapular retraction progressions	10 to 15 reps with 4-second eccentric contraction, 2-second isometric contraction at end-range, and 1-second concentric contraction
Integrate	Integrated dynamic movement	▪ Ball combo 1 ▪ Ball combo 2 ▪ Lunge to scaption ▪ Scaption progressions ▪ Squat to row	10 to 15 reps under control
*NOTE: Inhibit/lengthen bilaterally (right + left).			

Rest, Refuel, and Regenerate

Recovery Strategies



Recovery Questionnaire Targeted Responses

Rest	
Sleep	▪ Aim for 8+. ▪ Minimize sleep disruption beyond their baseline. ▪ Maximize the sense of feeling rested, which helps to gauge sleep quality. ▪ Minimize reliance on stimulants and empty calories, which helps to gauge sleep quality.
Relaxation	Target 60+ minutes per day.
Stress	▪ Reflect on client's perceived stress level. ▪ Track over time to assist the client in identifying behaviors and circumstances that correlate with stress levels.
Refuel	
Nutrition	Target a balanced nutritional approach appropriate to performance goals.
Hydration	▪ Aim to replace fluid lost through sweat and retain hydration status prior to the next bout of training. ▪ If less than 3% of body weight will be lost during competition or training, recommend fluid consumption as desired, or 500 mL (16.9 oz) before bed and within the hour prior to exercise.
Regenerate	
Pre-Activity	Warm-up and movement preparation should follow the Corrective Exercise Continuum.
Post-Activity	Cool-down and recovery or deloading workouts should follow the Corrective Exercise Continuum.

Rest, Refuel, and Regenerate Strategies

Phase	Strategies
Rest	▪ Aim for 8 hours of sleep per night. ▪ 60 minutes of accumulated psychological relaxation per day with activities such as breathing exercises, meditation, or reading ▪ Minimize the amount of perceived stress.
Refuel	▪ Choose nutritious meals/snacks higher in carbohydrates, moderate in protein, and low in fat. ▪ Modify macronutrient distribution to support activity goals and duration. ▪ Restore pre-activity hydration levels using pre- and post-activity weight measurements when possible.
Regenerate	▪ Use the Corrective Exercise Continuum to maximize movement quality and reduce muscle tension and overactivity.